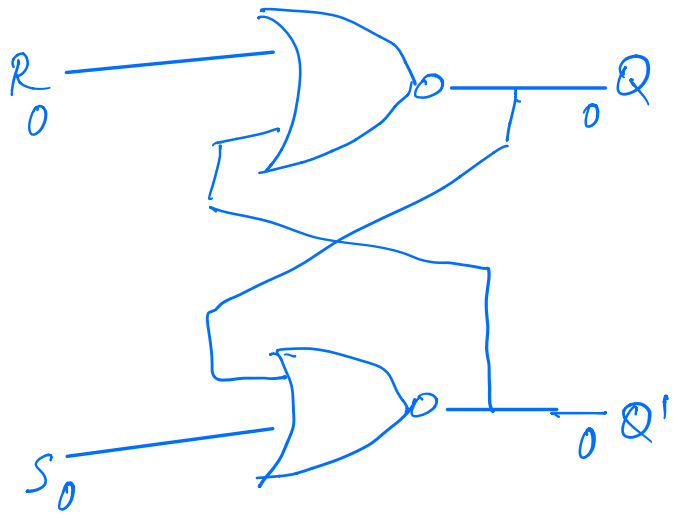
Why is $S=R=1$ an Invalid I/p in a SR/RS Latch and $S=R=0$, Invalid in the SR Latch with NAND gates ($S'R'$ Latch)? ?



$S=R=1$ drives $Q=Q'=0$

Subsequently i.e. when $S=R=0$ we have the following cases:

(i) The NOR gate with the quicker propagation would set its O/P to 1. If upper NOR is faster $Q=1, \bar{Q}=0$

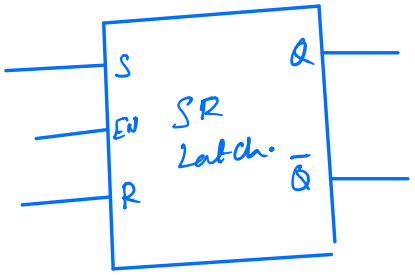
(ii) If lower NOR is faster $Q=0, \bar{Q}=1$.

(iii) ALSO LIKELY: Both Q and Q's zero reaches the upper and lower NOR gate at same time driving $Q=Q'=1$, and then leads

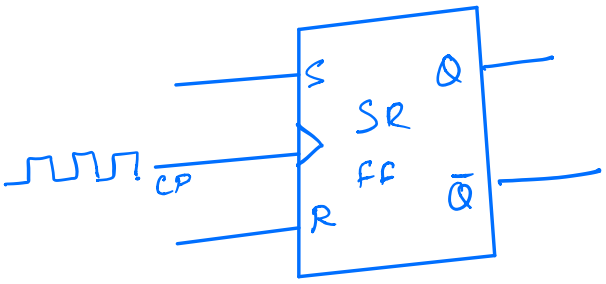
to toggling $Q = Q' = 1 \rightarrow Q = Q' = 0 \rightarrow Q' = Q' = 1 \rightarrow \ldots$

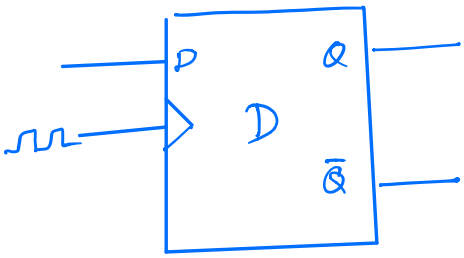
Flip Flops: When the latch is enabled or triggered with a CP then the latch is called a FF.

Symbols:

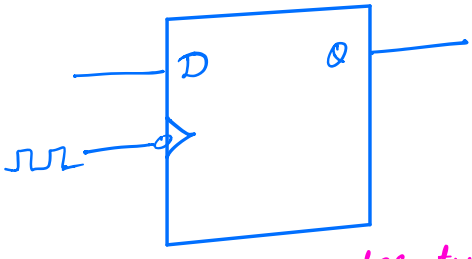


Q is the state of the latch. Q' as the name suggests is ALWAYS complement of Q.

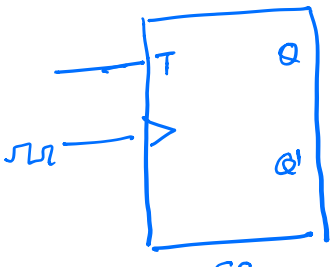




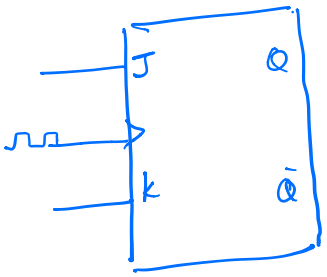
D FF +ve edge triggered



D FF -ve edge triggered



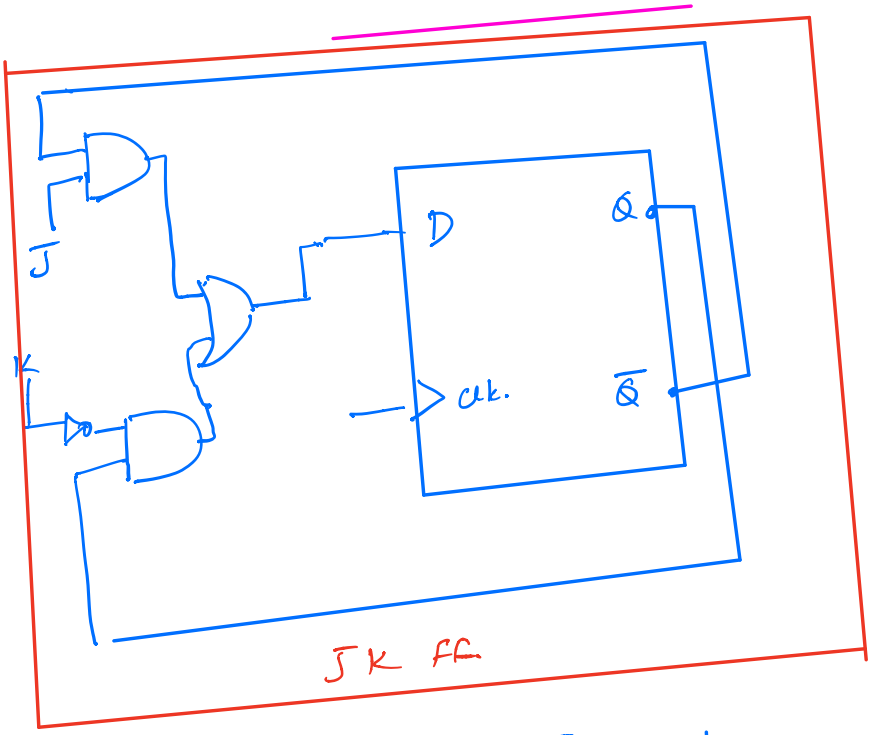
T FF



J FF.

Please note that the CP is always an IMPLICIT input to the flip flops but not shown as an explicit variable

JK FF:

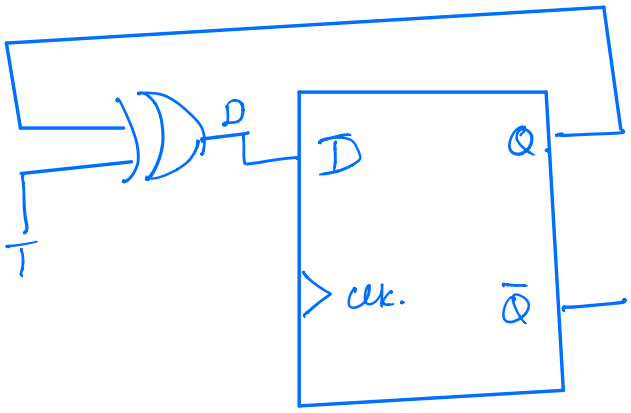


JK FF.

$D = J\bar{Q} + \bar{K}Q.$

J	K	Q(t)	$\bar{Q}$(t)	D	Q(t+1)	
0	0	0	1	0 Q(t)	0	Q(t) / no change.
0	0	1	0	1 Q(t)	1	
0	1	0	1	0	0	RESET
0	1	1	0	0	0	

1	0	0	1	1	1	SET
1	0	1	0	1	1	SET
1	1	0	1	1 $\bar{Q}(t)$	1	$\bar{Q}(t)$ or Toggle
1	1	1	0	0 $\bar{Q}(t)$	0	$\bar{Q}(t)$ or Toggle



$D = T\bar{Q} + \bar{T}Q$.

T	Q(t)	D	Q(t+1)	
0	0	0	0	Q(t) / No change
0	1	1	1	Q(t) / No change
1	0	1	1	$\bar{Q}(t)$ / TOGGLE.
1	1	0	0	$\bar{Q}(t)$ / TOGGLE.

FF Characteristic Table: Akin to Truth Table for combinational element.

J K

J	K	Q(t+1)
0	0	Q(t) (previous state / Hold)
0	1	0 (Reset)
1	0	1 (Set)
1	1	$\bar{Q}$(t) (Toggle)

D FF:

D	Q(t+1)
0	0 (Reset)
1	1 (Set)

T FF:

T	Q(t+1)
0	Q(t) (Hold)
1	$\bar{Q}$(t) (Toggle)